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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

Extra 3 - First Order Logic

Some of the following exercises are adapted from exercises in *"Logic in Computer Science: Modelling and Reasoning about Systems"*, by Michael R. A. Huth, Michael Huth and Mark Ryan.

3.1. Express the following propositions using logical connectives, quantifiers and boolean and arithmetic operations; consider as universe the set of integers $\mathbb{Z}$.

(a) The sum of two negative integers is negative.

(b) The difference between two negative integers is not necessarily negative.

(c) The sum of the squares of two integers is greater than or equal to the square of their sum.

(d) The modulus of the product of two integers is equal to the product of their moduli.

3.2. Let $\mathcal{L}$ be a language of first order with equality and such that $\mathcal{F}_0 = \{a, b\}$, $\mathcal{F}_1 = \{g\}$, $\mathcal{F}_2 = \{f, h\}$, $\mathcal{R}_1 = \{R, S\}$ and $\mathcal{R}_2 = \{P, Q\}$.

i. The length of a term t is the length of the sequence of characters that represent it. For example, $f(a, h(a, a))$ has length 11.

(a) Determine all ground terms of $\mathcal{L}$ with a length of less than 10 and in which the same functional symbol does not occurs more than once.

(b) Determine all ground terms of $\mathcal{L}$ with length less than 8.

ii. Indicate, **justifying**, which of the following expressions are terms of $\mathcal{L}$ and which are ground terms of $\mathcal{L}$:

(a) $h(a, f(a, g(a), g(a)))$

(b) $f(h(x, g(g(a))), x)$

(c) $f(a, P(a, g(x)))$

(d) $h(g(f(a, a)), f(b, a))$

(e) $f(h(x, h(y, y)), g(g(b)))$

(f) $f(a, g(h(g(x), x(a))))$

Laws of semantic equivalence

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \rightarrow \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	
$\neg\forall x\varphi$	$\Leftrightarrow$	$\exists x\neg\varphi$	(DeMorgan's Law)
$\neg\exists x\varphi$	$\Leftrightarrow$	$\forall x\neg\varphi$	(DeMorgan's Law)
$\forall x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\forall x\varphi \wedge \forall x\psi$	(but not with $\vee$)
$\exists x\varphi \vee \exists x\psi$	$\Leftrightarrow$	$\exists x(\varphi \vee \psi)$	(but not with $\wedge$)
$\forall x\forall y\varphi$	$\Leftrightarrow$	$\forall y\forall x\varphi$	
$\exists x\exists y\varphi$	$\Leftrightarrow$	$\exists y\exists x\varphi$	
$\forall x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\forall x\varphi \wedge \psi$	if x does not occur free in ψ
$\forall x(\varphi \vee \psi)$	$\Leftrightarrow$	$\forall x\varphi \vee \psi$	if x does not occur free in ψ
$\exists x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\exists x\varphi \wedge \psi$	if x does not occur free in ψ
$\exists x(\varphi \vee \psi)$	$\Leftrightarrow$	$\exists x\varphi \vee \psi$	if x does not occur free in ψ